



Same Feet, Shorter Walk

A Before/After Headcount of a Relocated Shuttle Stop's Ridership Bump

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The problem

Facilities relocated the East Precinct shuttle stop forty metres nearer the new engineering building midway through term, citing anticipated demand from the precinct's growing occupancy. Whether the resulting boarding increase is new patronage or displaced foot traffic has not been established.

Research questions

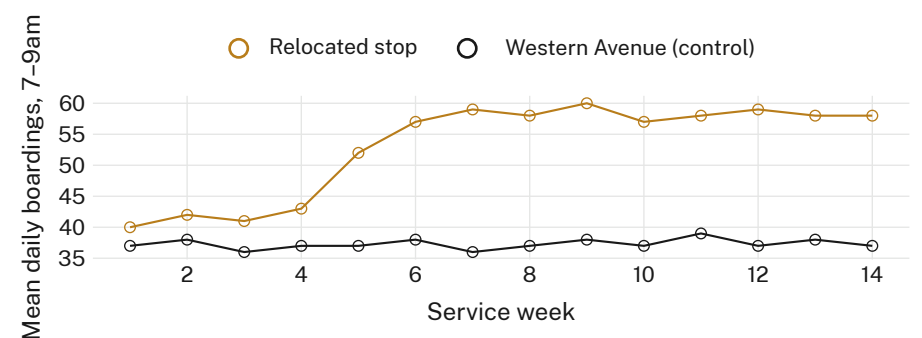
- Does the post-relocation boarding increase reflect new patronage or footfall already passing the new site?
- How does the boarding trajectory compare with an unmoved control stop over the same term?
- Does any gain outlast the relocation's first weeks of novelty?

Approach

- 14 weeks · manual 7–9am headcounts at both stops · $n = 3,014$ boarding events
- Difference-in-differences against the unmoved Western Avenue stop
- First 10 service days after the move excluded from the post-period mean, as a pre-registered novelty window

What we found

Weekday boardings at the relocated stop rose from a 4-week pre-move mean of 41.2 to a post-move mean of 58.4 per day, with the gain concentrated almost entirely in the first three service weeks. The Western Avenue control moved by less than a boarding a day across the same interval.



Relocated-stop boardings rose from a 4-week pre-move mean of 41.2 to a 58.4 post-move mean within three weeks of the 3 March move date, while the Western Avenue control held between 36.9 and 37.8 across the same interval.

Implications

The apparent ridership bump washes out almost entirely to a three-week reallocation of existing pedestrian traffic rather than new demand; the control stop's flat trajectory over the same interval argues against a broader precinct-wide shift. The pre-registered novelty window, intended to discard a transient settling-in effect, instead excludes the only interval in which the series moves at all — the boarding count is flat on either side of it. Results are consistent with a catchment that never grew, sampled forty metres further along the same desire line. The business case that funded the move forecast a standing increase, not a transient one. Next: extend the same difference-in-differences design to the two further stops relocated under the same precinct-growth business case.

Selected references

1. El-Geneidy, A., Grimsrud, M., Wasfi, R., Tétreault, P., & Surprenant-Legault, J. (2014). New evidence on walking distances to transit stops: identifying redundancies and gaps using variable service areas. *Transportation*, 41(1), 193–210. doi:10.1007/s11116-013-9508-z
2. Heinen, E., Panter, J., Mackett, R., & Ogilvie, D. (2015). Changes in mode of travel to work: a natural experimental study of new transport infrastructure. *International Journal of Behavioral Nutrition and Physical Activity*, 12, 81. doi:10.1186/s12966-015-0239-8
3. Sahu, P. K., Mehran, B., Mahapatra, S. P., et al. (2021). Spatial data analysis approach for network-wide consolidation of bus stop locations. *Public Transport*, 13, 375–394. doi:10.1007/s12469-021-00266-0
4. Borowski, E., Soria, J., Schofer, J., & Stathopoulos, A. (2022). Does ridesourcing respond to unplanned rail disruptions? A natural experiment analysis of mobility resilience and disparity. arXiv:2202.09466.

Headcounts recorded by hand tally at both stops; no boarding-pass or student ID data captured.

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